



Basilisk

MVP IMPLEMENTATION PLAN & COST MODEL

User Stories, Backlog, Cardano Data Strategy & Infrastructure Costs

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SECTION 1

1. What "MVP" Means

The PRD defines four phases. This plan covers **MVP = Phase 0 (Foundations) + Phase 1 (MVP Web)** — roughly **Months 0–5**. The goal: a usable web product at rough parity with TapTools' *core* analytics + portfolio surfaces, plus a public Data API beta.

In Scope

#	Surface	PRD ref
A	Portfolio Tracker (multi-wallet, ADA/USD, value-over-time, holdings, P&L)	§5.1 A
B	Token Charts (candles, OHLCV, live price, trades feed, token info)	§5.1 B
C	Market Screeners (top tokens, gainers/losers, trending, new)	§5.1 C
D	Wallet Profiler / Smart Money (any-address deep-dive, follow, whale feed)	§5.1 D
E	Token Distribution (top holders, concentration)	§5.1 E
F	Alerts (price / % move / whale / balance, multi-channel)	§5.1 F
G	Public Data API beta (REST, keys, free tier)	§5.1 G
—	The data engine under all of it: chain ingestion + DEX adapters + pricing engine	§7

Explicitly Deferred (NOT MVP)

- ✗ In-app swaps, WebSocket streaming API, full DeFi dashboard, NFT analytics suite, mobile apps, embeddable widgets.
- ✗ x402 agent API, MCP server, A2A trading rails — the **differentiators** come once the core is proven.

MVP SUCCESS BAR

A trader can connect/watch a wallet and see correct net worth + P&L; open any token and see a correct live price + chart + recent trades; screen the market; look up any wallet; set an alert that fires; and a developer can pull token/market/wallet data from a documented REST API with a free key.

2. Assumptions & Team

- **Team:** 2–4 engineers. Lean split: 1 backend/data (owns ingestion + pricing — the moat), 1 full-stack (API + portfolio/wallet), 1 frontend (web), + fractional design/PM. Estimates assume **~3 FTE**.
- **Estimates** are in **story points** (Fibonacci; 1 pt $\approx$ ½ day of one engineer incl. test + review). Planning numbers, not commitments.
- **Stack** (PRD §7.4): Next.js (web) · NestJS/TypeScript + Go for perf-critical ingestion · Postgres + TimescaleDB + Redis · Cardano data via **Demeter.run** (§6) · TradingView lightweight-charts.
- **Data strategy (§6):** start **managed** (Demeter + Blockfrost free fallback) rather than self-hosting a node — removes ~1–3 weeks of db-sync ops from the critical path.

SECTION 3

3. Epic Map & Milestones

```
M0 Foundations      → EPIC-0 Data Infra · EPIC-1 Platform/Auth · EPIC-9 DevOps
M1 Pricing+Charts   → EPIC-0 (DEX adapters, pricing) · EPIC-3 Token Charts
M2 Portfolio        → EPIC-2 Portfolio Tracker · EPIC-1 Wallet connect
M3 Discovery         → EPIC-4 Screeners · EPIC-5 Profiler/Smart-Money · EPIC-6 Dist
M4 Alerts + API      → EPIC-7 Alerts · EPIC-8 Public Data API beta
M5 Harden + launch  → perf · QA · docs · beta onboarding · observability
```

Epic	Title	Pts (est)
EPIC-0	Data Infrastructure: ingestion, DEX adapters, pricing engine	~110
EPIC-1	Platform: auth, wallet connect, API gateway/BFF	~45
EPIC-2	Portfolio Tracker	~55
EPIC-3	Token Charts	~45
EPIC-4	Market Screeners	~30
EPIC-5	Wallet Profiler / Smart Money	~40
EPIC-6	Token Distribution	~20
EPIC-7	Alerts	~40
EPIC-8	Public Data API (beta)	~35
EPIC-9	Web shell, design system, DevOps, observability	~50
Total		~470 pts

THE LONG POLE

At ~3 FTE, **~5 months is realistic** — consistent with the PRD's Month 0–5 window. **EPIC-0 is the critical path and the project's primary risk**; everything else depends on correct, fresh prices.

SECTION 4

4. Backlog: Epics → Stories → Tasks

Story IDs are `US-<epic>.<n>` . Each carries acceptance criteria (AC) and tasks. Priority: **MUST** **SHOULD** **NICE** .

EPIC-0 — Data Infrastructure (the moat)

Without correct, fresh prices nothing else matters. This epic is the critical path.

US-0.1 Connect to Cardano chain data **MUST**

8 pts

As the platform, I need reliable access to live + historical Cardano chain data so all downstream services have a source of truth.

AC: Read chain tip, stream new blocks/txs, query history; failover if primary degrades.

- Provision **Demeter.run**; stand up **Ogmios** + **Kupo** + **db-sync** (scale-to-zero) ports.
- Wire **Blockfrost free tier** (50k req/day) as REST fallback/cross-check.
- Thin internal `chain-data` client lib wrapping both, with health checks + circuit breaker.
- Define which addresses/policies Kupo indexes (DEX pool scripts + on-demand wallets).

US-0.2 Ingest the chain into our pipeline **MUST**

13 pts

As the data layer, I want to tail new blocks and persist relevant events for a real-time + queryable stream.

AC: New DEX swaps appear within the <5s freshness target; pipeline survives restarts and resumes.

- Ogmios/Oura chain-follower (Go/TS) → normalized event stream → Redis pub/sub.
- Idempotent checkpointing (resume from last slot) + Cardano rollback handling.
- Persist raw swap/pool events to TimescaleDB.

US-0.3 DEX adapters (decode pools & swaps) **MUST**

34 pts

As the pricing engine, I need each DEX's pools and swaps decoded into reserves + trade events to derive price and liquidity.

AC: Per DEX, reserves and swaps reconstructed accurately and validated vs the DEX's own UI for sample tokens.

- Adapter framework + shared interface (`getPools` , `decodeSwap` , `getReserves`).
- **Minswap** (do first — highest volume), **SundaeSwap**, **WingRiders**, **MuesliSwap**, **VyFinance** adapters.
- **Spike:** evaluate **Charli3 Dendrite** (OSS SDK already unifying these DEXs) to accelerate or replace hand-rolled adapters — could cut this story substantially.
- Per-adapter conformance tests + golden fixtures; alarm on decode failures (protocols upgrade scripts).

US-0.4 Pricing engine: VWAP + OHLCV **MUST**

21 pts

As a user, I want an accurate single price per token so charts, portfolio, and screeners all agree.

AC: Volume-weighted across pools (ADA base) → USD via oracle; OHLCV for 1m/5m/15m/1h/4h/1d; matches manual spot-check.

- VWAP aggregation across pools; **ADA/USD oracle** (cross-check ≥ 2 sources).
- OHLCV candle builder → TimescaleDB continuous aggregates; backfill top-N tokens via db-sync.
- Live candle updates → Redis pub/sub for chart/portfolio consumers.

US-0.5 Quality / spam filter **MUST**

8 pts

As a user, I don't want scam/dust tokens polluting screeners and prices.

AC: Tokens below liquidity/volume thresholds flagged "low quality" and excluded from default lists.

- Configurable thresholds (min liq, min vol, age); outlier handling for thin pools; quarantine list + admin override.

US-0.6 Token metadata service **MUST**

8 pts

As a user, I want token names, tickers, logos, decimals, supply, and policy IDs.

AC: Each token resolves to standard metadata (CIP-25/68 + registry); missing data degrades gracefully.

- Resolve from token registry + on-chain metadata; cache in Postgres; supply/FDV/mcap computation.

US-0.7 Reorg / data-correctness safeguards **MUST**

10 pts

- Rollback-aware writes; reconciliation job cross-checking prices vs Blockfrost; data-freshness SLA monitor (p95 dashboard); staleness alerting.

EPIC-1 — Platform: Auth, Wallet Connect, API Gateway

US-1.1 CIP-30 wallet connect (read-only) **MUST**

13 pts

As a user, I want to connect my Cardano wallet so the app can read my holdings.

AC: Connect Eternl, Lace, Nami, Vespr, Typhon via CIP-30; read-only; address(es) persisted; disconnect works.

- CIP-30 connector abstraction + per-wallet handling; multi-address support; watch-only address add (no wallet).

US-1.2 Account & auth (email/social + sign-in-with-wallet) **MUST**

13 pts

As a user, I want an account so my wallets, watchlists, and alerts persist.

AC: Sign up via email/social or CIP-30 sign-in-with-wallet; sessions; password reset; minimal PII.

- Auth service (consider Supabase Auth to save build time); session/JWT; wallet-signature login (CIP-8/30 `signData`).

US-1.3 API Gateway / BFF **MUST**

13 pts

As the system, I want one gateway handling auth, rate-limiting, caching, and API keys.

AC: All traffic flows through gateway; per-tier rate limits; response caching; request metering recorded.

- Gateway (NestJS); Redis rate limiter; API-key issuance/validation; per-endpoint cache TTLs; usage metering → Postgres.

US-1.4 Watchlists SHOULD

6 pts

As a user, I want to save tokens/wallets to a watchlist.

- Watchlist CRUD; sync across devices.

EPIC-2 — Portfolio Tracker

US-2.1 Net worth & holdings MUST

13 pts

As a holder, I want my total net worth and a holdings breakdown so I know what I own and its worth.

AC: Net worth in ADA + USD + chosen fiat; holdings across tokens, NFTs, LP (decoded), liquid-staking, lending; within freshness SLA.

- Portfolio service: UTxOs → balances (Kupo); decode LP to underlying; mark-to-market via pricing engine; multi-wallet aggregation; fiat conversion.

US-2.2 Value-over-time chart MUST

10 pts

As a holder, I want a chart of my portfolio value over time.

AC: Historical net-worth series (daily at MVP); handles wallets added later (backfill).

- Daily snapshot job + historical reconstruction from trade history; chart endpoint.

US-2.3 Realized / unrealized P&L MUST

13 pts

As a trader, I want P&L per position so I know if I'm up or down.

AC: Per-token realized + unrealized P&L from reconstructed trades; cost-basis method documented (e.g. avg cost).

- Trade reconstruction from on-chain swaps per wallet; cost-basis engine; edge cases (airdrops, LP, transfers).

US-2.4 Trade history SHOULD

8 pts

- Per-wallet chronological trade list: token, side, price, value, tx link.

US-2.5 Portfolio UI MUST

11 pts

- Portfolio dashboard; holdings table; allocation breakdown; value chart; P&L; empty/watch-only states.

EPIC-3 — Token Charts

US-3.1 Candlestick chart + OHLCV **MUST**

13 pts

As a trader, I want a fast candlestick chart with multiple timeframes.

AC: TradingView lightweight-charts; 1m→1d; sub-second load from cache; live last-candle updates.

- Chart component; OHLCV endpoint (Timescale continuous aggregates); live update via Redis subscription.

US-3.2 Live price + token info panel **MUST**

8 pts

AC: Live VWAP; supply, mcap, FDV, liquidity, holders, links, policy ID; price-change %, ATH/ATL.

- Token detail endpoint (pricing + metadata + distribution); info panel UI.

US-3.3 Recent trades feed **MUST**

8 pts

AC: Last ~500 trades (side, size, price, time, wallet) updating live.

- Trades endpoint + live feed UI; whale-trade highlighting.

US-3.4 Token page shell **MUST**

8 pts

- `/token/[id]` route; chart + info + trades + distribution tabs; share/deeplink; SEO basics.

US-3.5 Chart drawing tools / indicators **NICE**

8 pts

- Basic indicators (MA, volume); drawing tools (defer if tight).

EPIC-4 — Market Screeners

US-4.1 Top tokens & movers **MUST**

13 pts

As a trader, I want ranked lists (top by mcap/volume, gainers, losers, trending, new) to find opportunities.

AC: Ranked lists with quality thresholds; sortable/paginated; refreshes within SLA.

- Screener query layer over TimescaleDB; ranking jobs; new-listing detection; trending heuristic.

US-4.2 Filters **SHOULD**

8 pts

- Filter by liquidity, volume, age, category; saved filters.

US-4.3 Screener UI **MUST**

9 pts

- Screener tables; tabs; sparklines; responsive.

EPIC-5 — Wallet Profiler / Smart Money

US-5.1 Any-address profiler **MUST**

13 pts

As a user, I want to look up any wallet's total/liquid value, holdings, volume, activity, and P&L.

AC: Any address/handle → net worth, holdings, volume, activity, P&L; reuses portfolio engine read-only.

- Profiler endpoint (generalize portfolio service to any address); ADA Handle resolution; UI.

US-5.2 Smart-money lists & follow **MUST**

13 pts

As a trader, I want curated lists of profitable/high-activity wallets and to follow them.

AC: Curated lists (by PnL/volume/activity); follow a wallet; per-wallet activity feed.

- Wallet-scoring job (PnL/volume leaderboards); follow model; basic clustering/labeling.

US-5.3 Whale-trade feed **SHOULD**

8 pts

- Threshold-based whale-trade detection from swap stream → feed + (later) alert source.

US-5.4 Profiler / smart-money UI **MUST**

6 pts

- Wallet page; smart-money dashboard; follow buttons.

EPIC-6 — Token Distribution

US-6.1 Holder distribution **MUST**

13 pts

As a due-diligence user, I want to see holder concentration and top holders.

AC: Top holders, top-10 %, Gini, LP-vs-wallet split, supply distribution; excludes burn/known contracts.

- Holder enumeration per policy/asset (Kupo/db-sync); concentration metrics; known-address labeling; token-page UI.

EPIC-7 — Alerts

US-7.1 Alert rules engine **MUST**

13 pts

As a user, I want alerts on price thresholds, % moves, whale trades, and balance changes.

AC: CRUD alert rules; evaluated against live stream; fire-once/repeat; per-tier limits.

- Rule model + evaluation service consuming Redis pub/sub; dedupe/cooldown; persistence.

US-7.2 Multi-channel delivery **MUST**

13 pts

As a user, I want alerts via in-app, push, email, and Telegram/Discord.

AC: Delivery to in-app, web push, email, Telegram/Discord; retry on failure; channel prefs.

- Notification service; email (Resend/SES); web push; Telegram + Discord bots; delivery log + retries.

US-7.3 Alerts UI **MUST**

8 pts

- Alert creation from token/wallet pages; management page; in-app notification center.

US-7.4 New LP / new listing alerts **SHOULD**

6 pts

- Detect new pools/listings → alertable event.

EPIC-8 — Public Data API (beta)

US-8.1 REST endpoints (parity surface) **MUST**

13 pts

As a developer, I want REST endpoints for token, NFT(min), market, and wallet data.

AC: Endpoints mirroring PRD §2.3; stable, versioned (/v1), consistent schemas.

- Public API module over existing services; pagination; error contract; /v1 versioning.

US-8.2 API keys, tiers & rate limits **MUST**

10 pts

As a developer, I want a free key with clear rate limits.

AC: Self-serve key issuance; free tier + paid placeholders; usage dashboard; per-key metering.

- Key management UI; tier config; metering surfaced to user; quota enforcement at gateway.

US-8.3 Docs, SDK stub & sandbox **MUST**

12 pts

As a developer, I want great docs so I can integrate fast.

AC: OpenAPI spec; interactive docs; quickstart; TS client stub; example calls.

- OpenAPI generation; docs site (Scalar/Redoc); TS SDK stub; copy-paste examples.

EPIC-9 — Web Shell, Design System, DevOps & Observability

US-9.1 App shell & design system **MUST**

13 pts

- Next.js shell; navigation; theming (Cardano-blue / serpent-green brand); component library; light/dark; responsive; loading/empty/error states.

US-9.2 CI/CD & environments **MUST**

10 pts

- GitHub Actions (lint/test/build/deploy); dev/staging/prod; preview deploys; secrets in env/secret store (none in repo); IaC for backend.

US-9.3 Observability & error tracking **MUST**

10 pts

- Grafana Cloud (metrics/logs) + Sentry (errors); data-freshness SLA dashboard; uptime/alerting; structured logging.

US-9.4 QA, load test & beta onboarding**MUST**

10 pts

- E2E tests (Playwright) for core flows; load test pricing/API; beta waitlist + onboarding; feedback loop.

US-9.5 Security & privacy pass**MUST**

7 pts

- Authz review; rate-limit abuse tests; PII minimization; dependency scan; secrets audit before public launch.

5. Definition of Done

A story is **done** when: code reviewed + merged · unit/integration tests pass · meets AC · instrumented (logs/metrics) · no secrets in code · docs/changelog updated · deployed to staging and smoke-tested.

DATA STORIES CARRY AN EXTRA GATE

Any story touching prices, balances, or P&L additionally requires a **correctness cross-check against an external source** (Blockfrost / the DEX's own UI) before it's accepted.

6. Cardano Data Layer & the "Dedalus" Question

FINDING UP FRONT

"Dedalus" is not a real Cardano data-infra product. The name almost certainly refers to **Demeter.run** (by TxPipe) — the actual managed Cardano infrastructure PaaS (phonetically *De-meter* $\approx$ *De-dalus*). The only similar real thing is **Daedalus**, IOG's desktop full-node *wallet* — an end-user app, irrelevant to our data layer. **The intended product is Demeter, and it's a good fit.**

Options Evaluated

Provider	What it gives us	Pricing (Jun 2026)
Demeter.run ★	Hosted Ogmios, Kupo, Oura, db-sync, cardano-node — the exact primitives we'd otherwise self-host. Scale-to-zero.	Kupo ~\$0.09/100k req; Ogmios ~\$0.23/100k req; db-sync billed by connection-time, \$0 idle. Free tier (grant-subsidized). <i>Confirm exact rates in-dashboard.</i>
Blockfrost	Managed REST API (no streaming).	Free 50k req/day; Hobby €29/mo (300k/day); Developer €79/mo (1M/day).
Koios	Community decentralized REST API.	Free 50k/day; Pro ~\$30/mo (500k/day); Premium ~\$75/mo (1.2M/day).

Provider	What it gives us	Pricing (Jun 2026)
Maestro	Web3 stack incl. dedicated DEX/market-price endpoints — very on-topic.	Usage-based "compute credits"; has a free tier. <i>Exact USD pricing unverified (page JS-gated).</i>
Charli3 Dendrite	OSS SDK unifying DEX data (Minswap/Sundae/WingRiders/Muesli/VyFi/Spectrum/GeniusYield). <i>A library, not a host.</i>	Free (OSS).
Self-host node + db-sync	Full control, no per-req fees, full SQL.	Hetzner ~\$60–120/mo fixed; AWS ~\$350–550/mo. ~700 GB SSD, multi-day sync, ongoing ops.

RECOMMENDED PATH (MVP)

Demeter (Ogmios + Kupo + db-sync) as primary, Blockfrost free as fallback, Charli3 Dendrite to accelerate DEX parsing. Real-time prices via Ogmios/Oura + Dendrite-parsed pools; historical OHLCV via db-sync scale-to-zero SQL; wallet/portfolio UTxOs via Kupo. **Estimated Cardano-data cost at MVP scale: ~\$20–40/month.** Migration trigger: once historical-aggregation load climbs or we need guaranteed real-time latency at scale → move node + db-sync in-house on Hetzner (~\$60–120/mo fixed).

7. Infrastructure Cost Model (MVP)

Cardano data layer (\$6) is ~\$20-40/mo on top of the figures below. All prices June 2026; usage-billed items depend on real traffic. Hetzner excludes ~19% EU VAT.

Category	Scrappy (Hetzner)	Managed PaaS
Cardano data (\$6)	\$20-40 (Demeter + Blockfrost free)	\$20-40
Web hosting	\$0 (self-host on VM)	\$20 (Vercel Pro)
Backend compute (4-6 svcs)	\$25 (2× Hetzner + k3s)	\$40 (Railway/Fly)
Postgres	\$0 (on VM) or \$15 (DO)	\$25 (Supabase Pro)
Time-series DB (OHLCV)	\$0 (self-host Timescale)	\$30 (Timescale Cloud)
Redis (cache + pub/sub)	\$0 (on VM) or \$10 (Upstash)	\$10 (Upstash)
Monitoring + error tracking	\$0 (Grafana + Sentry free)	\$26 (Sentry Team)
Notifications (email/push/TG)	~\$2 (SES) + free channels	\$20 (Resend Pro)
Object storage (Cloudflare R2)	\$0-5	\$5
CDN (Cloudflare) / Domain / CI	~\$1	~\$5
TOTAL incl. data	≈ \$50-100 / mo	≈ \$200-225 / mo

\$50-100

Scrappy MVP / month
(self-managed on Hetzner)

\$150-180

Pragmatic hybrid most teams land on

\$200+

Fully managed PaaS / month

HOW IT SCALES

Time-series DB is the steepest curve — budget it as the #1 scaling cost. Keep storage/CDN on Cloudflare (R2 zero-egress) to avoid the AWS egress trap. Trajectory: MVP ~\$50–225/mo → 10k users ~\$500–1,500/mo → 50k+ users w/ heavy time-series \$2,000–6,000+/mo.

8. Key Risks & Mitigations

Risk	Impact	Mitigation
DEX decoder maintenance	Prices break silently	Conformance tests + golden fixtures + decode-failure alarms; lean on Charli3 Dendrite (community-maintained).
Data accuracy on thin pools	Wrong prices erode trust	Quality/outlier filters (US-0.5); cross-check vs Blockfrost (US-0.7).
Demeter cost/latency surprises	Budget or UX hit	Validate rates in-dashboard early; Blockfrost fallback; documented Hetzner escape hatch.
EPIC-0 on critical path	Slips whole MVP	Start Day 1; spike Charli3 Dendrite early; Minswap-first end-to-end.
Scope creep into Phase 2	MVP never ships	Differentiators explicitly deferred (\$1); hold the line.
P&L correctness	Misleading P&L	Document cost-basis method; reconcile vs known wallets; explicit edge cases.

SECTION 9

9. Immediate Next Actions (First 2 Weeks)

1. **Stand up Demeter** (Ogmios + Kupo + db-sync); confirm live pricing in-dashboard. *(US-0.1)*
 2. **Spike Charli3 Dendrite** vs hand-rolled adapters — decide approach before committing 34 pts. *(US-0.3 spike)*
 3. **Minswap end-to-end vertical slice**: ingest → decode → VWAP → one token's live price + candle on a bare page. Proves the whole pipeline. *(US-0.2 → US-0.4)*
 4. **Pick ADA/USD oracle** + provision Timescale/Postgres/Redis. *(US-0.4)*
 5. **Scaffold app shell + CI/CD + Sentry/Grafana** so feature work lands in a real pipeline from day one. *(EPIC-9)*
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